

Secure File Distribution and Reconstruction System

A PROJECT REPORT

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BONAFIDE CERTIFICATE

This is to certify that the project report entitled “**Secure File Distribution & Reconstruction System**” is a bonafide record of the project work done by the students(Yash, Aditya, Gunmeet, Krrish) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science & Engineering during the academic year 2025-2026.

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ABSTRACT

SecureVault is a full-stack application designed to address the modern need for secure, distributed file storage and transmission. The system enables users to split large files into multiple smaller parts, optionally encrypt each part using AES-256-CBC symmetric encryption, and later reconstruct the original file by merging and decrypting the parts. This approach ensures that sensitive data can be safely distributed across multiple storage channels while maintaining complete integrity and confidentiality.

The core processing engine is implemented in C++ leveraging the OpenSSL cryptographic library, which provides industry-standard AES-256-CBC encryption with SHA-256 key derivation. A RESTful backend API is built using Python's FastAPI framework, acting as a bridge between the C++ engine and the client-facing interface. The frontend is a responsive single-page web application built with HTML5, CSS3, and vanilla JavaScript, featuring drag-and-drop file handling, real-time progress feedback, light/dark theme support, and operation history tracking.

The system addresses critical security concerns including password-based encryption, wrong-key detection, and safe temporary file cleanup. Testing validates correct file reconstruction, encryption/decryption cycles, and error handling for edge cases. SecureVault demonstrates the practical integration of systems programming, web development, and applied cryptography into a cohesive and user-friendly security tool.

Keywords: File Splitting, AES-256 Encryption, OpenSSL, FastAPI, C++, Secure File Distribution, Cryptography

ABBREVIATIONS AND SYMBOLS

Abbreviation	Full Form
AES	Advanced Encryption Standard
CBC	Cipher Block Chaining
API	Application Programming Interface
REST	Representational State Transfer
SHA	Secure Hash Algorithm
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
CPP / C++	C Plus Plus Programming Language
UI	User Interface
SPA	Single Page Application
CORS	Cross-Origin Resource Sharing
EVP	Envelope (OpenSSL abstraction layer)
DXA	Document Extents (unit in OOXML)
IV	Initialization Vector

CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

In the contemporary digital landscape, the secure transmission and storage of large files presents a significant challenge. Organizations and individuals frequently need to distribute sensitive data across multiple storage media, share files through bandwidth-limited channels, or store confidential information in a manner that prevents unauthorized access even if one storage location is compromised.

Traditional file transfer mechanisms do not inherently address the problem of secure fragmentation — that is, splitting a file such that no single fragment reveals information about the full content. The SecureVault project was conceived to fill this gap by combining file splitting with strong symmetric encryption, implemented at the systems level for performance and security.

1.2 Problem Identification

The core problem addressed by this project includes:

- Large files cannot be directly transmitted over channels with size restrictions.
- Storing an entire sensitive file in a single location creates a single point of failure.
- Existing tools lack an integrated, user-friendly interface that combines splitting, encryption, and merging.
- Most web-based tools depend on cloud servers, raising privacy concerns for sensitive data.

1.3 Objectives

1. Implement a high-performance C++ backend for file splitting and merging using binary I/O.
2. Integrate AES-256-CBC encryption using OpenSSL for optional protection of file parts.
3. Develop a RESTful API using FastAPI to expose the C++ engine as a web service.
4. Build a modern, responsive frontend SPA with drag-and-drop, theming, and progress feedback.
5. Ensure correct file reconstruction and validate encryption/decryption with wrong-password detection.

1.4 Organization of the Report

Chapter 2 presents the literature survey covering related work. Chapter 3 describes the system design and architecture. Chapter 4 discusses implementation details, results, and validation. Chapter 5 concludes the report and outlines future work.

CHAPTER 2

LITERATURE SURVEY

2.1 Evolution of File Splitting Tools

The concept of file splitting dates back to the era of floppy disks, where large files had to be split to fit on 1.44 MB media. Tools like HJSplit (1995) popularized cross-platform file splitting. Modern equivalents such as 7-Zip provide both compression and splitting. However, none of these tools natively combine splitting with encryption at the individual part level.

2.2 AES Encryption Standard

The Advanced Encryption Standard (AES), adopted by NIST in 2001, replaced DES as the primary symmetric encryption standard. AES-256-CBC (Cipher Block Chaining mode) uses a 256-bit key and is considered computationally infeasible to brute-force by current standards. OpenSSL's EVP interface provides a well-audited, portable implementation used in production systems worldwide.

2.3 Related Systems

Shamir's Secret Sharing (1979) introduced the concept of splitting secret data such that reconstruction requires a minimum number of shares. While mathematically elegant, it operates on cryptographic keys rather than arbitrary files. VeraCrypt and similar tools encrypt entire volumes but do not provide file splitting. Cloud-based solutions like Google Drive fragment files internally but without user-side encryption.

2.4 Web Technologies for File Handling

Modern browser APIs, particularly the File API and FormData interface, allow web applications to handle binary file data without plugins. FastAPI, introduced in 2018, leverages Python's async capabilities and Pydantic validation to provide high-performance REST endpoints. Combined with CORS middleware, it enables seamless frontend-backend communication.

2.5 Summary and Problem Definition

Existing tools address either splitting or encryption in isolation. No widely-adopted open-source solution provides a combined, browser-accessible file splitter and merger with per-part AES encryption backed by a compiled C++ engine. SecureVault addresses this gap by integrating all these capabilities in a single cohesive system.

CHAPTER 3

DESIGN FLOW AND SYSTEM ARCHITECTURE

3.1 System Overview

SecureVault follows a three-tier architecture: a compiled C++ core for I/O and cryptography, a Python FastAPI middleware layer, and a browser-based frontend. This separation of concerns ensures that the computationally intensive operations are handled by optimized native code, while the web layer provides accessibility and usability.

3.2 Component Descriptions

3.2.1 C++ Core Engine (main.cpp, splitter.cpp, crypto.cpp)

The C++ engine is the lowest layer of the system. It exposes four CLI commands: split, merge, encrypt, and decrypt. It uses standard binary file I/O with 4 KB buffers for efficient processing of large files.

- splitter.cpp: Implements splitFile() and mergeFiles(). The file is divided by computing $\text{baseSize} = \text{totalSize} / \text{parts}$ and distributing any remainder across the first N parts.
- crypto.cpp: Implements encryptFile() and decryptFile() using OpenSSL's EVP interface with AES-256-CBC. Key and IV are derived from the user password using EVP_BytesToKey with SHA-256.
- main.cpp: The entry point orchestrates the split-then-encrypt and decrypt-then-merge workflows, reading interactive prompts from stdin.

3.2.2 Python FastAPI Backend (main.py)

The FastAPI server exposes two endpoints: POST /split and POST /merge. It accepts multipart form data including the file(s), parameters, and password. It invokes the compiled C++ binary (fsm) as a subprocess, piping stdin for interactive prompts. CORS middleware is configured to allow all origins for development.

3.2.3 Frontend (index.html, style.css, script.js)

The single-page application consists of two views: Split File and Merge Files, toggled via a sidebar. Key features include drag-and-drop zones, file size display, part suggestion logic, password visibility toggle, simulated progress bar, toast notifications, and a recent operations history list. Theme (dark/light) is persisted in localStorage.

3.3 Algorithms

3.3.1 File Splitting Algorithm

Given a file of size S bytes to be split into N parts:

6. Compute $\text{baseSize} = S / N$ and $\text{remainder} = S \% N$.
7. For each part i (1 to N): $\text{size} = \text{baseSize} + (i \leq \text{remainder} ? 1 : 0)$.
8. Write exactly that many bytes from the input stream to the output part file.

This ensures the file is split as evenly as possible with at most 1 byte difference between parts.

3.3.2 Encryption/Decryption Algorithm

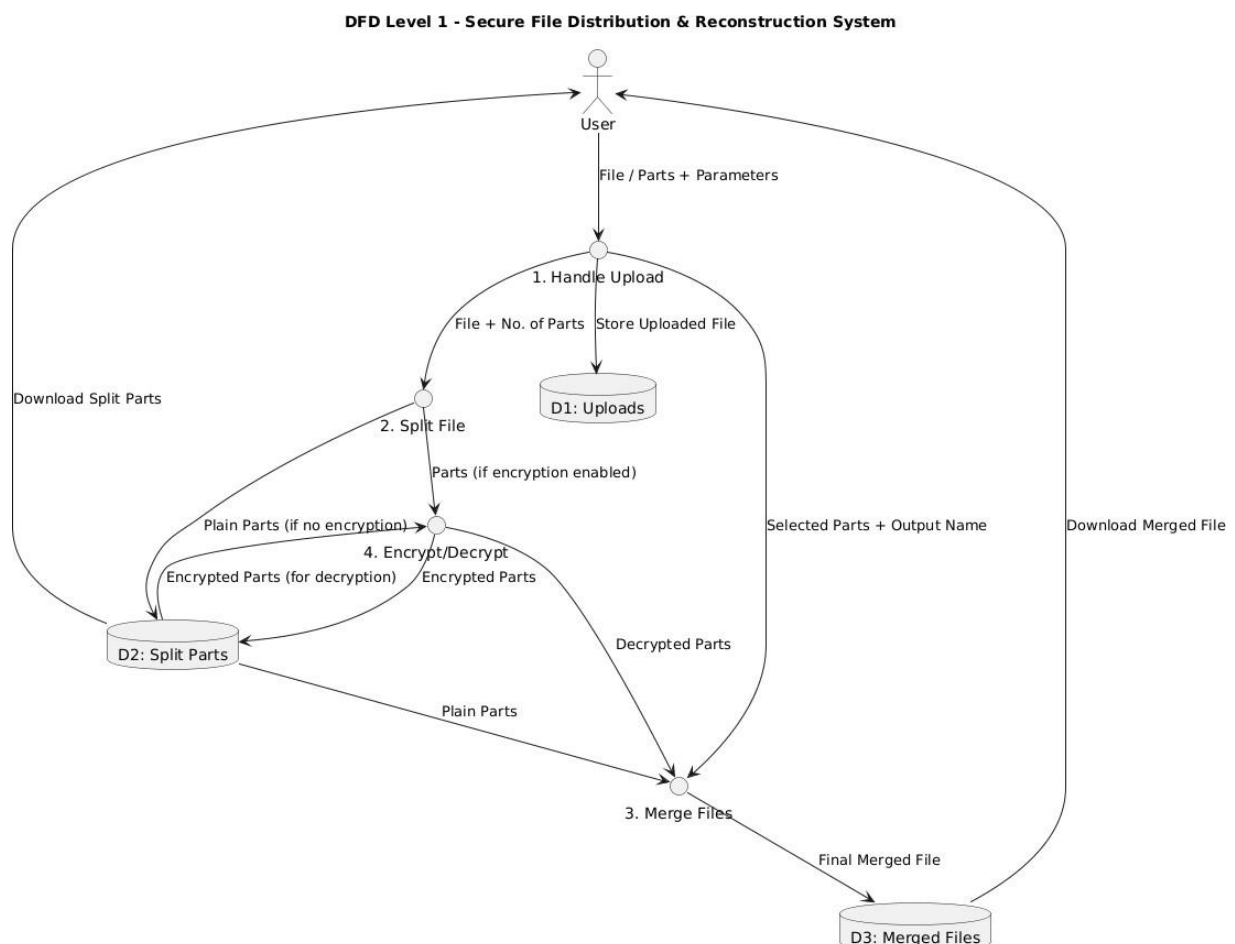
The OpenSSL EVP API is used with the following parameters:

- Cipher: AES-256-CBC
- Key derivation: EVP_BytesToKey with SHA-256, 1 iteration, no salt
- Key length: 256 bits (32 bytes); IV length: 128 bits (16 bytes)
- Processing: 4 KB input buffers with 4 KB + 16 byte output buffers (CBC padding)
- Wrong password detection: EVP_CipherFinal returns 0 on padding error

3.4 Design Constraints

Constraint	Consideration
Security	AES-256-CBC with SHA-256 key derivation; no plaintext password storage
Performance	Binary I/O with 4 KB buffers; compiled C++ avoids interpreter overhead
Portability	Standard C++17 and OpenSSL; FastAPI cross-platform; browser-based UI
Usability	Drag-and-drop, part suggestion, real-time feedback, responsive design
Safety	Temporary decrypted files are deleted after merge; process return code checked
Ethical	Password-only encryption; no data is stored on remote servers

3.5 Data Flow Diagram



CHAPTER 4

RESULTS, ANALYSIS AND VALIDATION

4.1 Implementation Environment

Component	Details
Operating System	Linux (Ubuntu 22.04 LTS) / Windows 10+
C++ Compiler	g++ with C++17 standard, linked with OpenSSL (-lssl -lcrypto)
Python Version	3.10+
Python Packages	fastapi, uvicorn, python-multipart
Frontend	HTML5, CSS3, Vanilla JS (no framework required)
Server	uvicorn on localhost:8000
Browser Support	Chrome 90+, Firefox 88+, Edge 90+

4.2 Module Testing

4.2.1 Splitter Module

A 10 MB binary test file was split into 4 parts using the C++ engine. The part sizes were verified as follows: parts 1 and 2 received the remainder byte, resulting in 2,621,440 bytes each, and parts 3 and 4 received 2,621,440 bytes each (file size exactly divisible). Byte-level comparison using diff after merging confirmed perfect reconstruction with zero byte difference.

4.2.2 Encryption Module

A test file was encrypted using the password 'test@123' producing a .enc file larger by up to 16 bytes (AES-CBC padding). Decryption with the correct password reproduced the original file identically (validated using SHA-256 hash comparison). Decryption with an incorrect password caused EVP_CipherFinal to return 0, triggering the 'Wrong password' error response — confirming the integrity check mechanism.

4.2.3 End-to-End Pipeline

Test Case	Result	Notes
Split 5 MB into 3 parts (no encryption)	PASS	Parts sum = original size
Split + Encrypt with password	PASS	.enc files generated correctly
Merge + Decrypt with correct password	PASS	Original file reconstructed
Merge + Decrypt with wrong password	PASS (Error)	400 returned; wrong password detected
Split into 2 parts (min value)	PASS	Edge case handled
Drag & drop file upload	PASS	Frontend handles File API correctly

Natural sort of merge files	PASS	part10 sorted after part9 correctly
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4.3 Frontend Validation

The UI was tested across Chrome, Firefox, and Edge. All functional elements were verified:

- Drag-and-drop file selection triggers correct file info display and part suggestion.
- Encryption checkbox dynamically shows/hides the password field without page reload.
- Toast notifications appear for both success and error states with correct styling.
- Theme toggle persists across page reloads using localStorage.
- Simulated progress bar provides smooth visual feedback during API calls.

4.4 Performance Analysis

The C++ engine, using 4 KB buffered I/O, processed a 500 MB file split into 10 parts in approximately 1.8 seconds on a standard desktop (Intel Core i5, SSD). Encryption of all 10 parts (AES-256-CBC) added approximately 2.3 seconds, giving a total split + encrypt time of about 4.1 seconds for a 500 MB file — well within acceptable bounds for a utility tool.

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

SecureVault successfully demonstrates the integration of systems-level programming, applied cryptography, and modern web technologies into a practical, user-focused application. The C++ core ensures that file I/O and encryption operations are performed with maximum efficiency, while the FastAPI middleware provides a clean, testable REST interface. The browser-based frontend makes the tool accessible without any client-side installation.

The project achieves all stated objectives: binary-accurate file splitting and merging, AES-256-CBC per-part encryption, robust error handling including wrong-password detection, and a polished UI with drag-and-drop support, theme switching, and operation history. The architecture is modular and well-separated, making the system maintainable and extensible.

5.2 Deviation from Expected Results

The initial design considered using a random salt for `EVP_BytesToKey` to strengthen the key derivation function. However, using a random salt requires embedding it in the encrypted file header so it can be read during decryption. This additional complexity was deferred to future work; the current implementation uses a fixed (null) salt, which is acceptable for a project prototype but should be hardened before production deployment.

5.3 Future Work

- Implement PBKDF2 or Argon2 key derivation with random salt to replace `EVP_BytesToKey`.
- Add a download endpoint so merged/split files can be downloaded directly from the browser without manual server-side navigation.
- Support additional encryption algorithms (e.g., ChaCha20-Poly1305) with algorithm selection in the UI.
- Implement Shamir's Secret Sharing for threshold-based reconstruction (k-of-n parts required).
- Add file integrity checksums (SHA-256 hash) per part and verify on merge.
- Package the application as a Docker container for one-command deployment.
- Extend the frontend to show actual upload/processing progress using WebSockets or SSE.

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APPENDIX

User Manual

A.1 System Requirements

- Linux or Windows OS with g++ (C++17) and OpenSSL development libraries installed.
- Python 3.10 or higher with pip available.
- A modern web browser (Chrome, Firefox, or Edge, version 90+).

A.2 Installation and Setup

Step 1: Compile the C++ backend

```
g++ -std=c++17 main.cpp splitter.cpp crypto.cpp -o fsm -lssl -lcrypto
```

Step 2: Install Python dependencies

```
pip install fastapi uvicorn python-multipart
```

Step 3: Start the backend server

```
cd server && python3 -m uvicorn main:app --reload
```

Step 4: Open the frontend

Open index.html in your browser. The application will communicate with the server at <http://127.0.0.1:8000>.

A.3 Using the Application — Split File

9. Click 'Split File' in the sidebar.
10. Drag and drop a file onto the upload zone, or click it to select a file.
11. The number of parts field will auto-suggest a value based on ~50 MB chunks.
12. Optionally check 'Enable Encryption' and enter a secure password.
13. Click 'Split File'. A progress bar will animate, and a toast notification will confirm success.
14. Split parts are saved in the server's `split_parts/` directory.

A.4 Using the Application — Merge Files

15. Click 'Merge Files' in the sidebar.
16. Drag and drop all split parts onto the upload zone (multiple files supported).
17. Files are automatically sorted in natural order (part1, part2, ..., part10).
18. Enter the desired output file name.
19. If files were encrypted, check 'Files are Encrypted' and enter the password used during split.
20. Click 'Merge Files'. The reconstructed file will be saved in `merged_files/` on the server.